

Remote Health Monitoring System for the Elderly

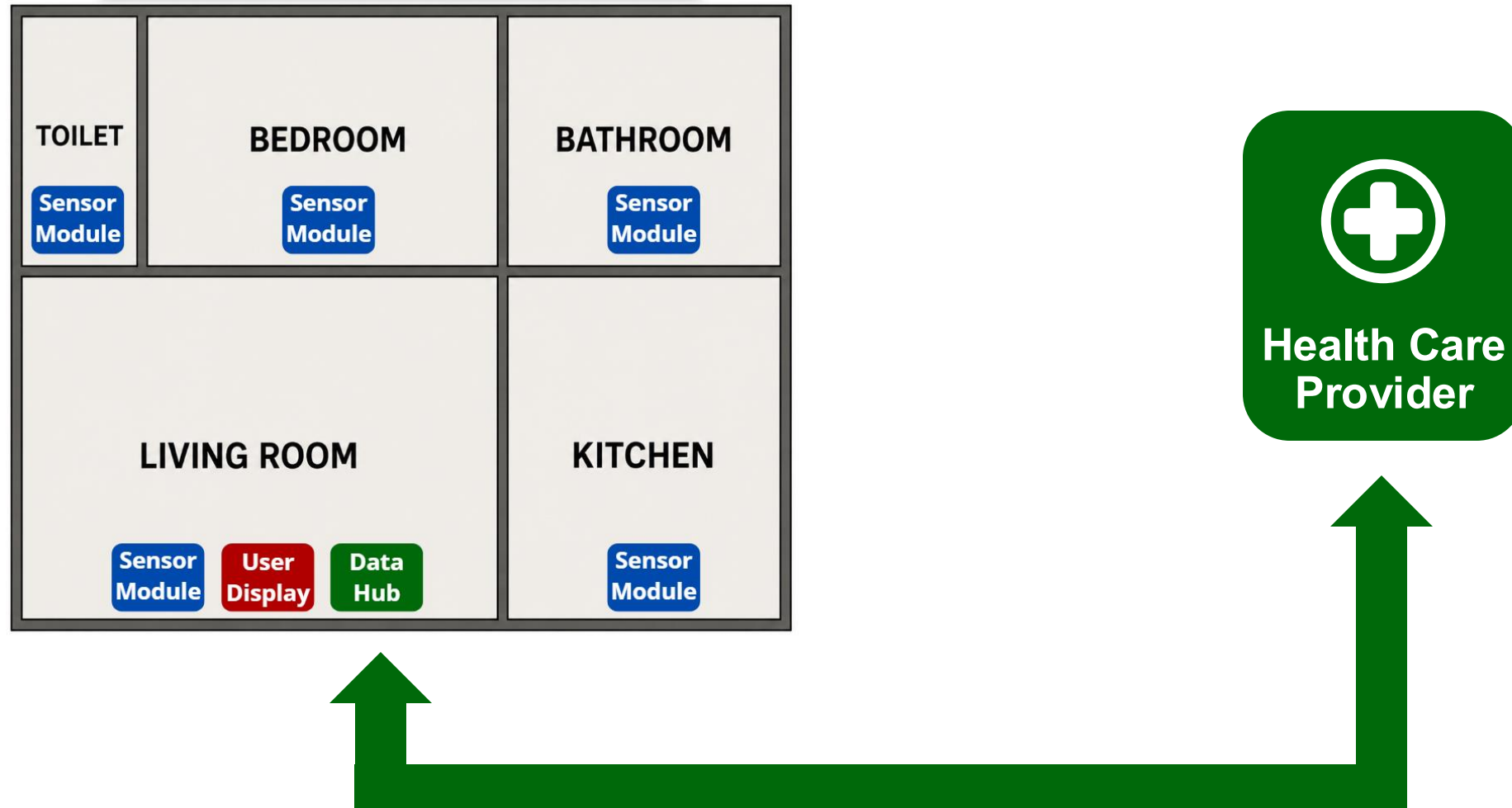
Yasmina Aarkoub · Mohamed Almifalani · Frederik Schiepers

7-7-2026

An ageing population, a widening care gap

- The over-65 group is growing faster than the workforce that supports it
- Increase of elderly living alone
- Risks: accidents, medical complications, decease
- Remote Monitoring of vital signs, environmental signs and activity
 - Fast response: detection of emergencies
 - Prevention: detect changes over time which indicate health complication using pattern recognition
 - Reduction of burden on healthcare resources

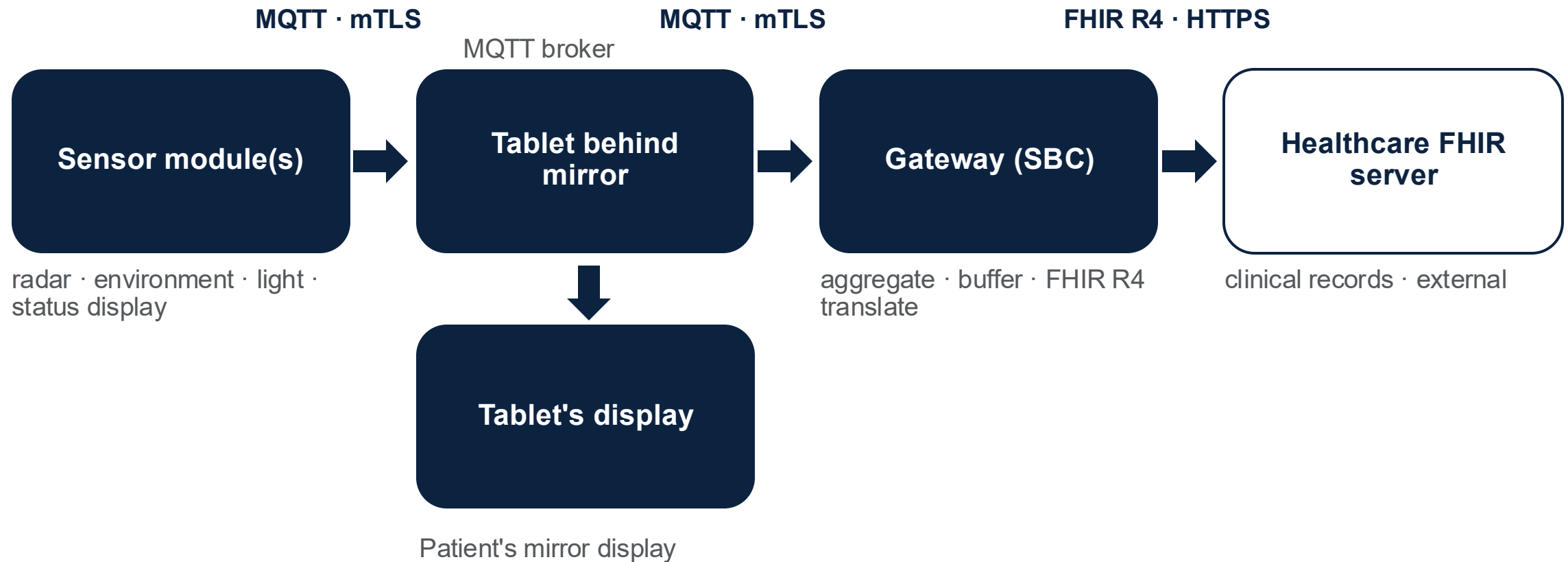
Solution: Remote Health Monitoring System



Goal and requirements

- Observe presence, vital signs, and environment at home, with no wearable, or user input
- Radar measures heart rate and respiratory rate
- Design a modular PCB
- Display activates on presence and shows vitals in an elderly-friendly way
- Security: devices authenticate each other before any data flows, traffic encrypted mutual TLS 1.2 · per-device certificates (sensor module ↔ MQTT broker)
- Interoperability: vital signs as HL7 FHIR R4 observations with LOINC codes over HTTPS

System architecture



Why millimetre-wave radar

Conclusions from literature:

Pearson coefficients: 0.95, 0.97 for HR and BR respectively.[1]

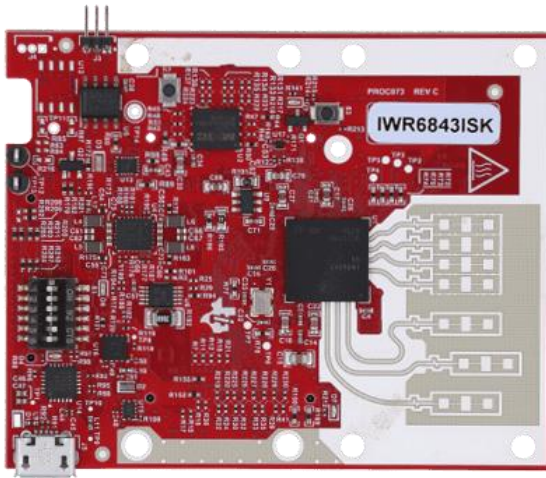
Detection accuracies greater than 93% for both vitals.[2]

Error below 7%.[3]

Reported high HR accuracy after measuring against real time ECG signal.[4]

[1] D. Wang et al. | [2] Y. Wang et al. | [3] Z. Hao et al. | [4] M. A. Nasser & L. Gromert

Radar module selection



Texas Instruments: TIWR6843



Infineon: BGT60TR13C



•Acconeer: A121 – Pulsed Coherent Radar:

Require heavy signal processing

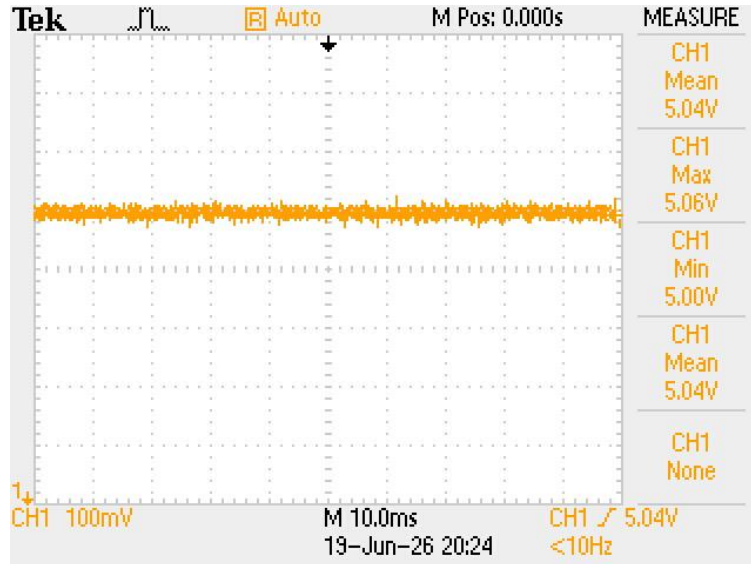
Radar module selection

- **Vital signs:** MR60BHA2
 - Outputs processed heart-rate and breathing-rate values
 - Faster integration than raw radar platforms
- **Presence detection:** HMMD
 - Low-cost presence radar
 - Suitable for wake-up / activation logic

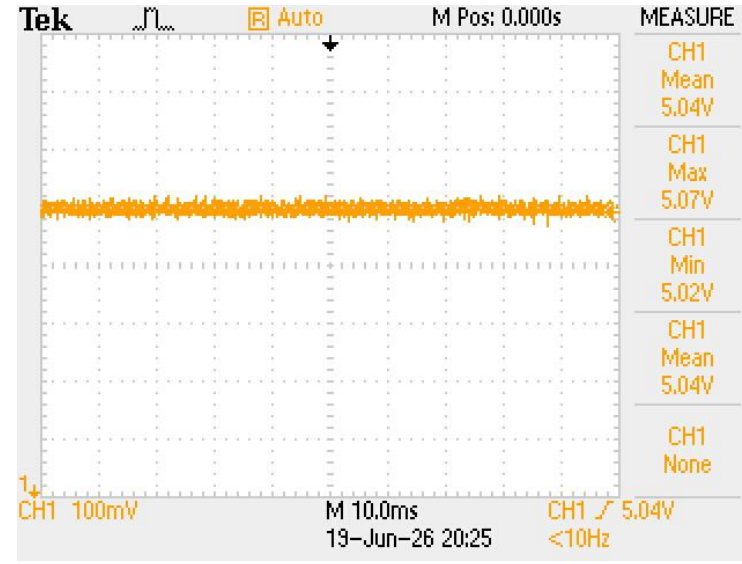
PCB Design Goal

- **Goal:**
 - One compact modular PCB for sensing, display, power, and control.
- **Integrated:**
 - Raspberry Pi Pico 2 WH
 - Environmental sensors
 - MR60BHA2 / C1001 / HMMD radar
 - Support LDR light sensor
 - OLED display
 - Buttons and status LEDs
 - 12 V (via Buck-Converter) / 5 V power options

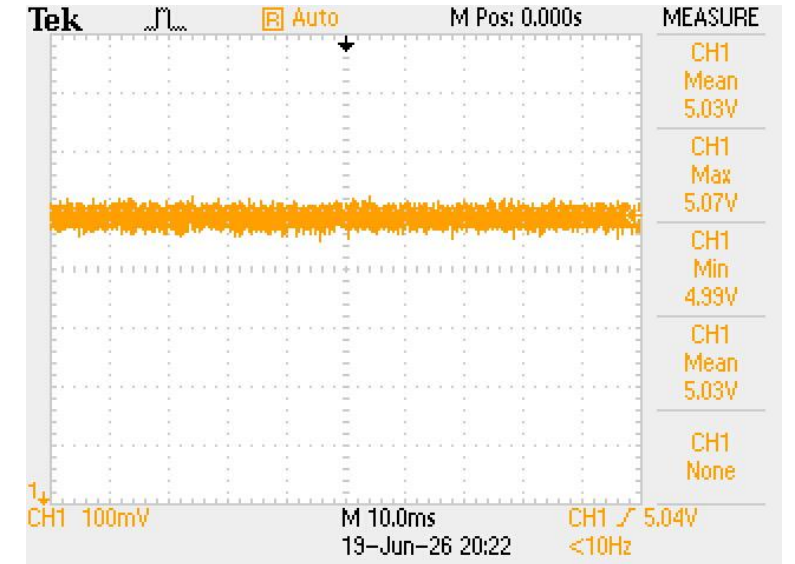
PCB-Buck Converter



No-load



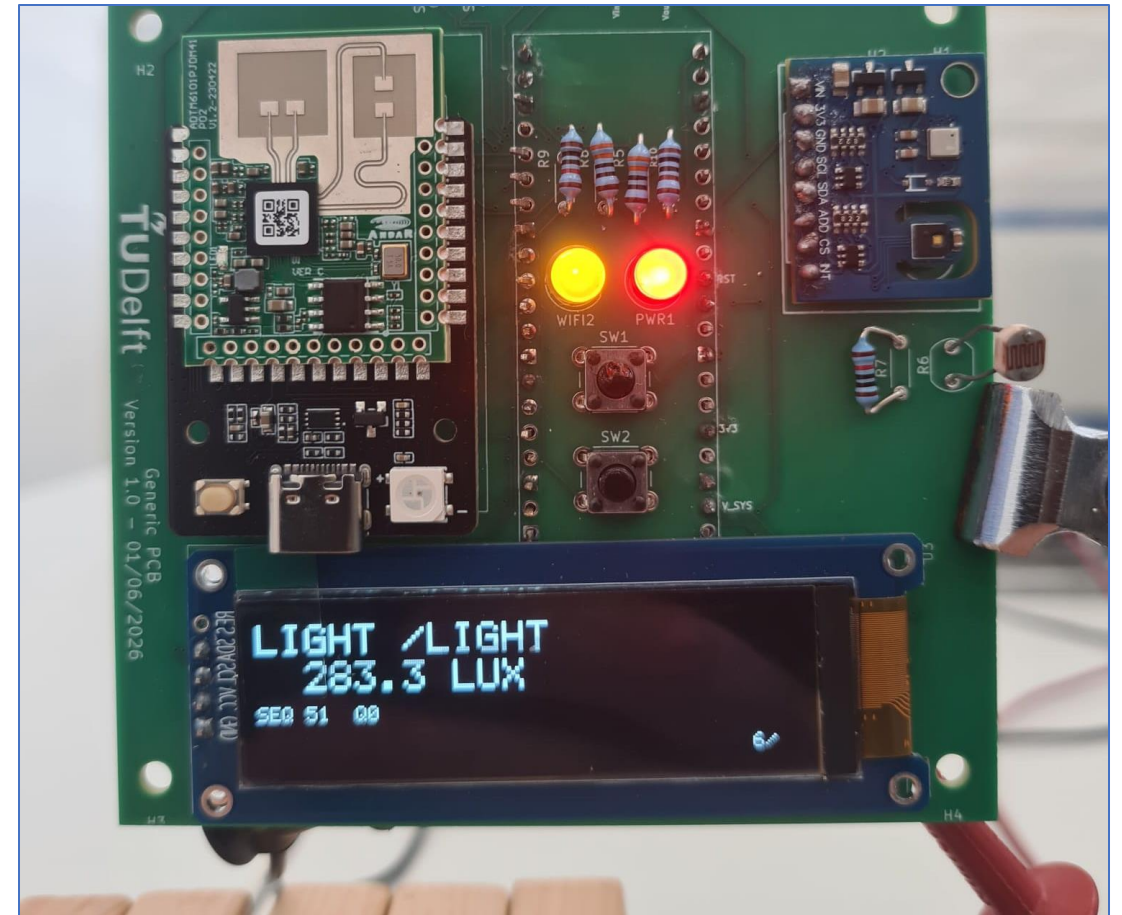
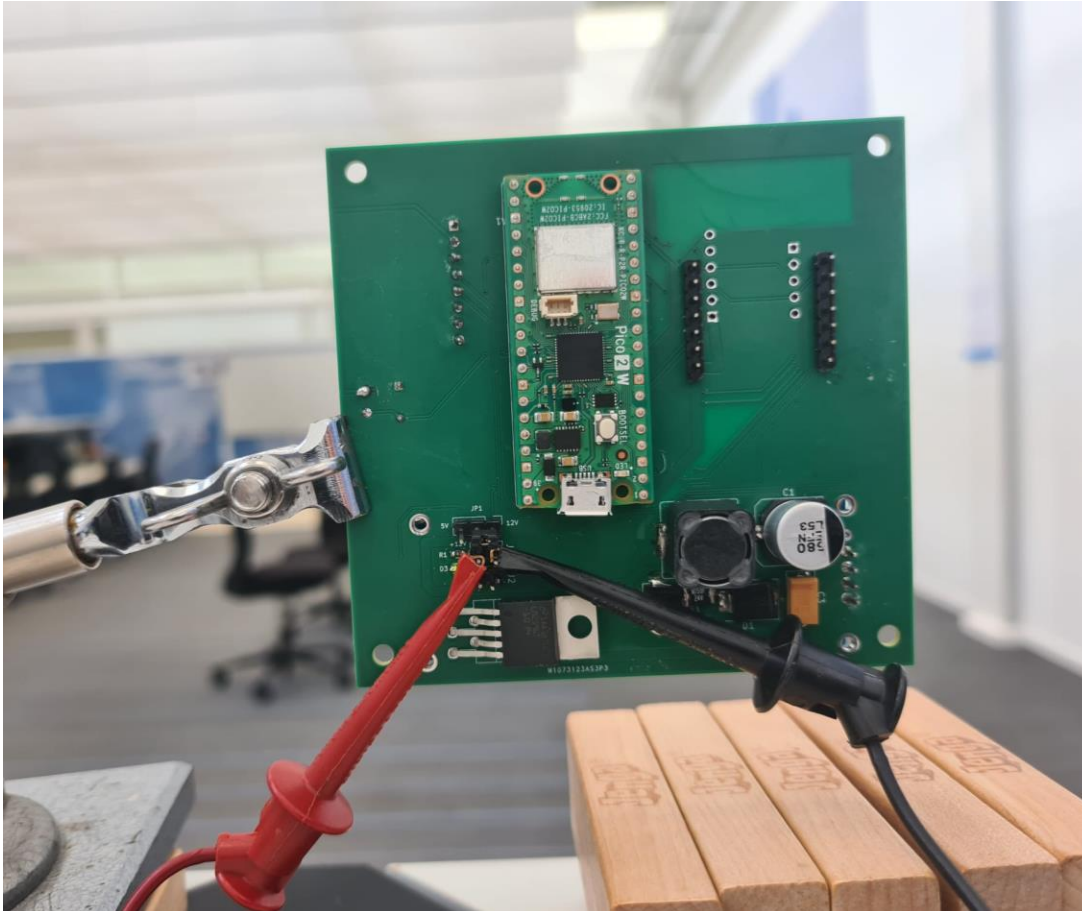
220 Ohm load



10 Ohm load

Maximum peak-to-peak variation = 80 mV

PCB- Photos



Improve reliability of pre-processed MR60BHA2

- **Goal:**
 - Improve reliability of processed radar output.



Trade-off: more stable output, but increased latency

HMMD Presence Filtering

- **Goal:**
 - Avoid false wake-up decisions.

OT2 detects
presence

UART confirm
presence

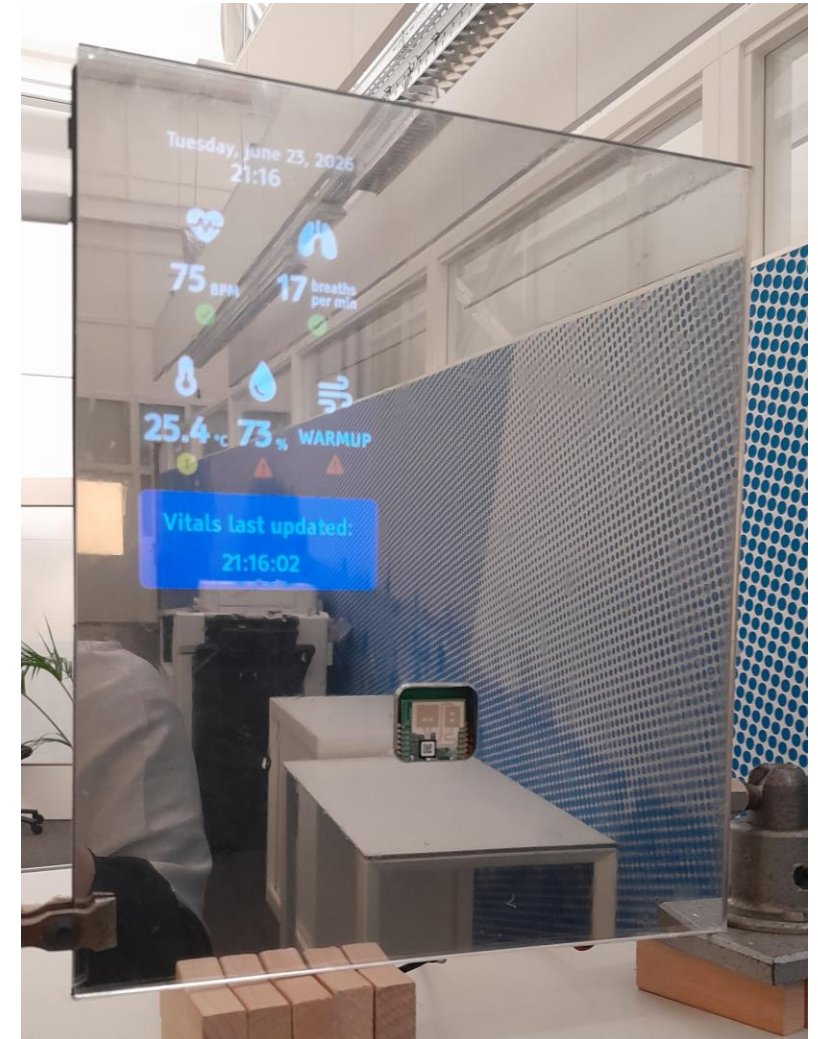
Gate 0 ignored as
noisy

Distance
smoothed with
EMA

Activation only
between 90-150
cm

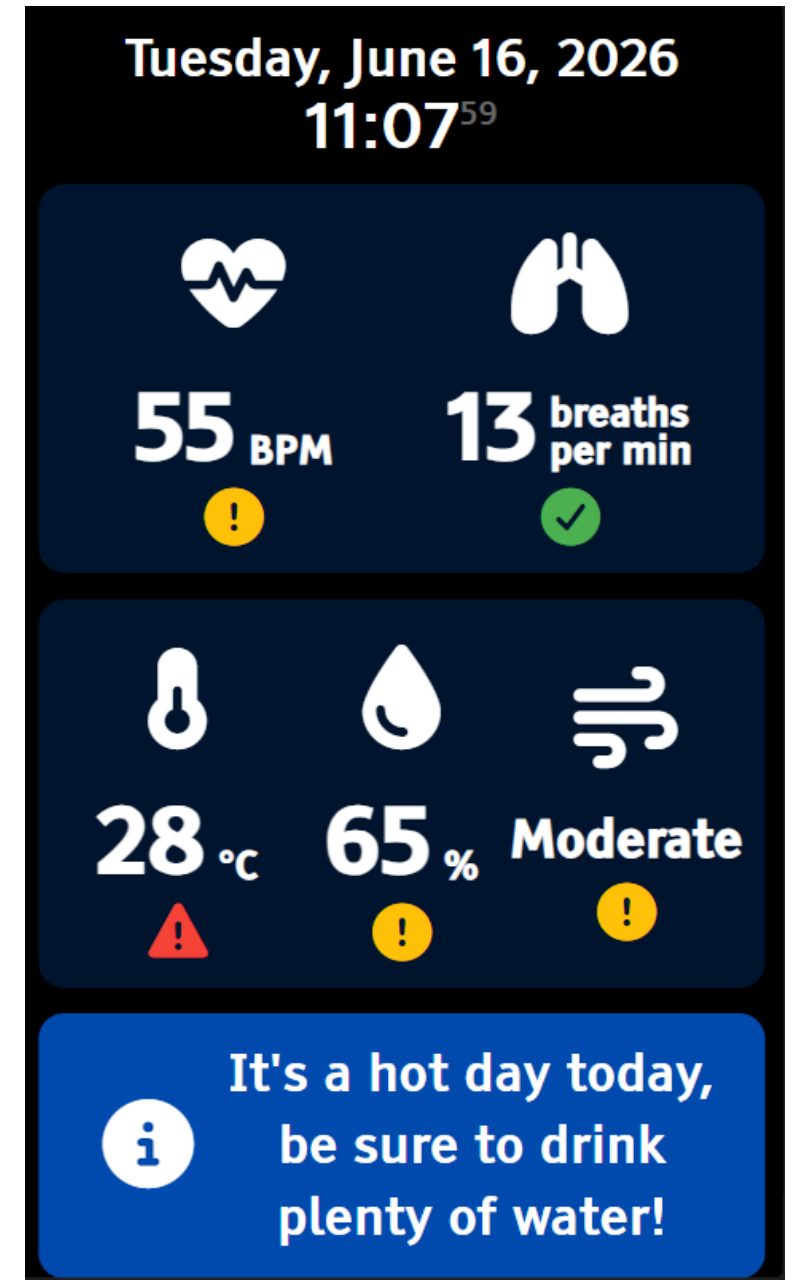
The Smart Mirror: information without interaction

- A tablet behind a two-way mirror
 - an ordinary mirror when idle
 - a health display on approach
- (De)activates on presence, with no input required



The Smart Mirror: UI Designed for Elderly

- Large icons and text
- Font for users with impaired vision
- High contrast
- Limited visual information
- Traffic-light scheme to interpret data



The Smart Mirror: Use of an Android Tablet

- Why: High functionality at low cost
- How:
 - MQTT message broker and UI platform both Linux based
 - Termux: Terminal emulator for Android
 - Custom application for remote screen (de)activation
 - Add-ons exist to start the system automatically at boot

Conclusion

- A working concept demonstrator: an end-to-end path from home sensing to an easy to read user interface
- Modular sensor PCB
Radar follows general trend. However, accuracy is questionable.
- Several technical domains integrated into one architecture, with a redesigned static mTLS security model
- Tablet provisioning: one script installs and configures the Mosquitto broker (packages, mTLS certs, ACLs) reproducibly
- The Program of Requirements was met to a large extent at concept-demonstrator level
- Not a certified medical device, but a step toward safer independent living

Future work

- Move to a raw-data radar platform and build a tailored processing pipeline
- Complete security testing, production endpoint authentication, and on-device buffering
- Add resident identification via face recognition for personalised mirror content
- Further research in long-term reliability of hardware

Demo

Thank you. Questions?

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